

Douglas Alencar

DATA & AI ENGINEER

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portfolio-douglas-alencar.web.app

PROFESSIONAL SUMMARY

Data & AI Engineer with 7 years of experience in technology, combining data engineering, analytics, governance, and AI application development. Proven track record in demanding corporate environments across healthcare, industry, and applied research, focused on turning data and models into usable, traceable, production-ready solutions. Hands-on experience with Python, SQL, PostgreSQL, GCP, BigQuery, FastAPI, and generative AI applications using RAG, embeddings, semantic search, and LLM integration.

CORE COMPETENCIES

AI Applications RAG, embeddings, semantic search, LLMs, agents, and evaluation.	Data Engineering Python, SQL, PostgreSQL, BigQuery, pipelines, and cloud.	Backend & APIs FastAPI, Pydantic, SQLAlchemy, REST, and Docker.
Cloud & Analytics GCP, Azure, Fabric, Power BI, and enterprise platforms.	AI-assisted Engineering SDD, harness, Codex, Claude Code, Git, testing, and CI/CD.	Governance & Production Architecture, security, observability, quality, and deployment.

PROFESSIONAL EXPERIENCE

- A.C.Camargo Cancer Center** | Data, Analytics, Governance & AI **2 years**
 - Managed and coordinated data, analytics, and AI initiatives in a highly complex oncology healthcare environment.
 - Led data governance, connecting business needs, data quality, architecture, and responsible data use.
 - Worked with the GCP data ecosystem, including BigQuery, pipelines, processing services, and data governance capabilities.
- AB InBev** | Technology & Global Corporate Environment **2.5 years**
 - Worked in IT within a large-scale corporate environment, strengthening experience with data, enterprise systems, processes, and cross-functional collaboration.
 - Worked in an international, execution-driven environment, communicating with diverse technical and business stakeholders.
- Shell + UFRJ** | Engineering, Technology & Applied Research **2.5 years**
 - Worked in a university-industry partnership combining engineering, technology, and applied research in a multidisciplinary environment.
 - Developed experience in structured problem solving, technical documentation, and collaboration with multiple stakeholders.

EDUCATION & INTERNATIONAL EXPERIENCE

Postgraduate Studies in Engineering - UFRJ / UFRRJ • **Science Without Borders** - 18-month international exchange • Advanced English

TECHNICAL STACK

Data: Python • SQL • PostgreSQL • BigQuery • Spark/PySpark • dbt • Pandas • Polars
AI: RAG • Embeddings • Semantic Search • LLM Applications • AI Agents • Structured Outputs
Backend: FastAPI • Pydantic • SQLAlchemy • REST • Alembic • Docker
Cloud & Analytics: GCP • Azure • Fabric • Power BI • SAP
Engineering: Git • SDD • Harness • Codex • Claude Code • Testing • CI/CD • Observability

SELECTED PROJECTS

AI Customer Service & Scheduling Platform | Public case study in production

2026

Demonstration AI customer service and scheduling application, built as a technical portfolio project and deployed to production. The domain and data are fully simulated; no real patients or real-world personal data are used.

- Combines RAG, LLMs, relational and vector databases, customer service workflows, and multiple levels of autonomy.
- Modular monolith architecture. Backend: Python 3.11, FastAPI, Pydantic, SQLAlchemy 2.x, and Alembic; frontend: React, TypeScript, and Vite.
- Production deployment on GCP using Cloud Run, Cloud Build, Artifact Registry, Secret Manager, and Firebase; PostgreSQL hosted on Neon.
- First functional version delivered in 15 days with Codex and Claude Code support, following SDD: 69 commits, 11 SDD packages, 473 tasks, 28 migrations, 247 tests, 18 E2E smoke scripts, and 17 Playwright scenarios.
- Engineering lifecycle: constitution → specify → clarify → plan → tasks → analyze → implement → converge, with decisions and limitations documented.
- LangChain and LangGraph were intentionally not used in this version; LLM integration, retrieval, and orchestration were implemented directly in the application.

Live application: plataforma-atendimento-prod.web.app

Data Engineering on GCP | Portfolio project

In development

- Project designed to demonstrate pipelines, processing, modeling, data quality, and data delivery on Google Cloud.
- Complements the AI case study and reinforces the hybrid Data & AI Engineer positioning.

Data & AI Engineer Portfolio | Personal product

In development

- Web resume built as an engineering product: static architecture, light/dark mode, case study pages, and low-friction contact options.
- Built with Next.js, React, TypeScript, Tailwind CSS, Firebase Hosting, and CI/CD on GitHub.
- No backend in V1 by architectural decision: static export, content decoupled from presentation, and structure prepared for PT/EN support.

PROFESSIONAL DIFFERENTIATORS

- Combines corporate data/governance experience with end-to-end AI application development.
- Able to engage with business stakeholders, design architecture, and drive implementation through testing and production.
- Uses coding agents within a controlled engineering process with specifications, traceability, and validation - not just ad hoc code generation.

LEARN MORE

Portfolio: portfolio-douglas-alencar.web.app | **GitHub:** github.com/alencardoug

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